THE DELTA-INVARIANT DOMINATES THE NUMBER OF GEOMETRIC BRANCHES MINUS ONE FOR STRICTLY HENSELIAN ONE-DIMENSIONAL REDUCED NAGATA LOCAL RINGS

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Allowed standard prerequisites (editorial).

- Finiteness over a henselian ring and characterization of discrete valuation rings. Stacks Project, Algebra, Lemma algebra-lemma-finite-over-henselian (tag 0DYD) and Lemma algebra-lemma-characterize-dvr (tag 034Z). The first gives that the normalization decomposes as a product of local henselian rings and the second identifies the normal local factors as discrete valuation rings.
- The delta-invariant as the length of the quotient by the normalization. Stacks Project, Varieties, Lemma lemma-prepare-delta-invariant (tag 09NP). Identifies the delta-invariant with the length of the quotient, which is what the inequality compares with the number of branches.

Source reference. Original reference 0C26: sources/stacks-chow/tags.txt, source lines 15146–15146. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0C3S: sources/stacks-chow/tags.txt, source lines 15202–15202. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0C37: sources/stacks-chow/tags.txt, source lines 15182–15182. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04GH: sources/stacks-chow/tags.txt, source lines 5833–5833. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00PD: sources/stacks-chow/tags.txt, source lines 927–927. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(A, \mathfrak{m})$ be a strictly henselian 1-dimensional reduced Nagata local ring. Then*

$$\delta\text{-invariant of } A \geq \text{number of geometric branches of } A - 1$$

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If equality holds, then A is a wedge of $n \geq 1$ strictly henselian discrete valuation rings.

Proof. The number of geometric branches is equal to the number of branches of A (immediate from More on Algebra, Definition 0C26). Let $A \to A'$ be as in Lemma 0C3S. Observe that the number of branches of A is the number of maximal ideals of A', see More on Algebra, Lemma 0C37. There is a surjection

$$A'/A \longrightarrow \left(\prod\nolimits_{\mathfrak{m}'} \kappa(\mathfrak{m}')\right)/\kappa(\mathfrak{m})$$

Since $\dim_{\kappa(\mathfrak{m})} \prod \kappa(\mathfrak{m}')$ is $\geq$ the number of branches, the inequality is obvious.

If equality holds, then $\kappa(\mathfrak{m}') = \kappa(\mathfrak{m})$ for all $\mathfrak{m}' \subset A'$ and the displayed arrow above is an isomorphism. Since A is henselian and $A \to A'$ is finite, we see that A' is a product of local henselian rings, see Algebra, Lemma 04GH. The factors are the local rings $A'_{\mathfrak{m}'}$ and as A' is normal, these factors are discrete valuation rings (Algebra, Lemma 00PD). Since the displayed arrow is an isomorphism we see that A is indeed the wedge of these local rings. □